MUMUCD: A Multimodal Multiclass Change Detection Dataset

F. Serva, A. Sebastianelli, *Member, IEEE*, B. Le Saux, *Senior Member, IEEE*, and F. Ricciuti

***Abstract*—This work introduces the multimodal multiclass change detection (MUMUCD) dataset, which comprises 70 globally distributed georeferenced bitemporal pairs obtained by multiple spaceborne sensors. Acquisitions over heterogeneous terrains (e.g., urban, rural, forests, deserts) for the period 2019–2024 are processed to create the first large-scale curated dataset combining synthetic aperture radar (SAR, Sentinel-1), multispectral (MS, Sentinel-2), and hyperspectral (HS, PRISMA) data with ancillary information. Provided with a resampled image resolution of 10 m and a size of 1536 $\times$ 1536 square pixels, MUMUCD allows to extract several thousands of nonoverlapping patches with size 128 $\times$ 128 square pixels, enabling data-intensive machine learning (ML) applications. Seasonality plays a critical role in the analysis of environmental data, so we carefully selected scenes representing all times of the year and a variety of geographical contexts relevant to key impact sectors, taking also into account the availability of PRISMA acquisitions. While scenes with low cloudiness are prioritized, cloudy pixels are not excluded as different data combinations (e.g., SAR/optical) can be exploited to mitigate the atmospheric effects. Beyond coregistered data, we include surface elevation, land cover (LC), and binary change maps obtained by processing the Dynamic World (DW) dataset, based on the provided MS data. Benchmarking of machine and deep learning (DL) algorithms indicates that the provided labels should be augmented to get the most out of the multiple modalities. MUMUCD is meant to fill a research gap by offering multisensor and task-oriented data, which make it ideal to fine-tune standard and foundational ML models for a variety of tasks, and even unique when these tasks involve CD or HS measurements.**

***Index Terms*—Change detection (CD), deep learning (DL), hyperspectral (HS) data, multimodal dataset.**

I. Introduction

CHANGE detection (CD) approaches are widely used with remotely sensed images, starting more than four decades ago with seminal studies by Malila [1] who focused on vegetation status. The availability of multispectral (MS) sensing instruments increased the range of possible applications of CD methods, as the information obtained by combining multiple bands can provide deeper insights on surface conditions. One limitation of supervised methods is the need for ground-truth (GT) data, which can be strongly application-specific and generally require either ground measurements (which is sparse) or manual annotation. The latter is moreover resource-intensive and prone to error, especially when applied to large, multimodal archives. Since reliable GT data are not always available, significant attention has been given in the past years [2] to the use of machine and deep learning (ML and DL), given their generalization capabilities upon a sufficiently complete training step.

Initially used mostly for surface characterization, for example, mineral mapping and material detection, hyperspectral (HS) data from spaceborne sensors are nowadays becoming increasingly available [3]. HS measurements improve upon the spectral resolution by a factor of 10 or more, enabling a number of downstream tasks, including CD applications [4]. While some CD studies relying on HS data appeared in the past few years [5], [6], [7], they have generally used a limited number of images or resorted to unsupervised approaches [8].

With this work, we aim to provide a large-scale objectively annotated dataset including multimodal remote sensing data, and in particular highly resolved spectral data, which are not yet fully exploited also due to the lack of open datasets. With respect to the state-of-the-art, our contribution offers multisensor capacity [9] and supports CD applications [10], thanks to multitemporality, providing enhanced flexibility. The adoption of a common and easy-to-use format allows us to provide an analysis-ready dataset (ARD), which we believe will help users tackling the problem of sparse annotation for land cover (LC) CD applications.

II. Methods and Data

A. Data Collection and Preprocessing

Our new multimodal multiclass change detection (MUMUCD) dataset combines data from different spaceborne sensors with ancillary data to provide a global and diverse archive usable for multiple applications.

Synthetic aperture radar (SAR) data are obtained from the Sentinel-1 (S1) mission [11] of European Space Agency (ESA). The mission includes multiple satellites: S1A launched in 2014 and still operational, S1B operating between 2016 and 2021, and S1C launched in late 2024. Acquisitions from the first two satellites are considered for MUMUCD, with a useful revisit time depending on the location but better than two weeks (at least until the decommissioning of S1B). The S1 C-band radar has all-weather capabilities, but given the strong dependence on sensing geometry, in MUMUCD we select data taken along the same relative orbit for the ground range detected (GRD) mode, with a spatial resolution of 5 $\times$ 20 m^2. Backscattering in two polarizations (VV and VH, or

Received 6 May 2025; revised 20 June 2025; accepted 30 June 2025. Date of publication 3 July 2025; date of current version 17 July 2025. *(F. Serva and A. Sebastianelli contributed equally to this work.) (Corresponding author: F. Serva.)*

F. Serva is with the Institute of Marine Sciences, National Research Council, 00133 Rome, Italy, and also with the Φ-Lab, European Space Agency, 00044 Frascati, Italy (e-mail: federico.serva@terrarum.eu).

A. Sebastianelli with the Φ-Lab, European Space Agency, Frascati 00044, Italy.

B. Le Saux is with DG CNECT, European Commission, 1049 Bruxelles, Belgium, and also with the Φ-Lab, European Space Agency, 00044 Frascati, Italy.

F. Ricciuti resides in Milan, Italy.

Digital Object Identifier 10.1109/LGRS.2025.3585797

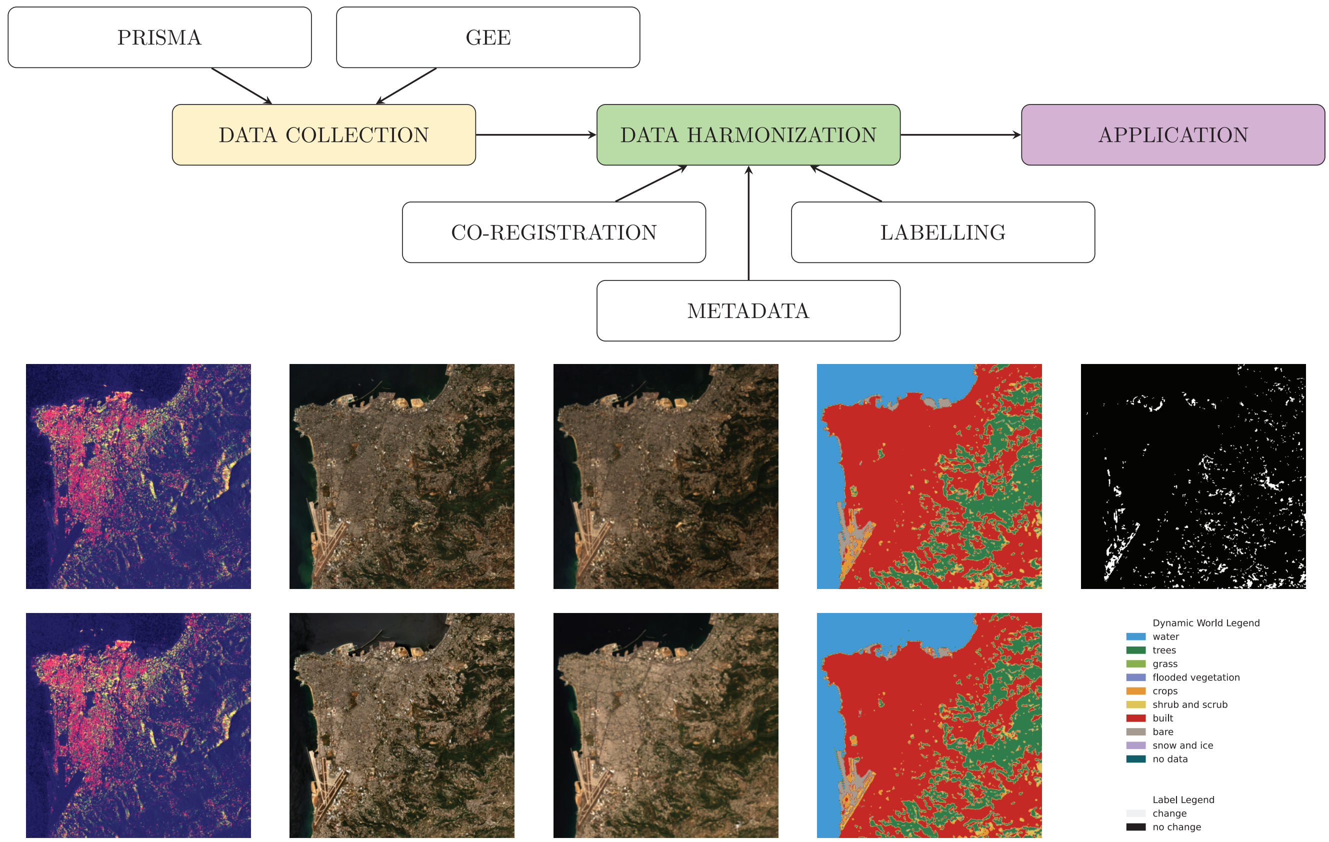

Fig. 1. (Top) Workflow of the procedure used to prepare the MUMUCD dataset. The bottom panel reports before (top row) and after (bottom row) images for a selected scene, with (from left to right) an S1 VV-VH composite, S2 and PRISMA true color composites, DW, and the binary change map.

HV and HH for high-latitude areas) and incidence angle are provided.

The Sentinel-2 (S2) mission [12] provides MS data in the visible-near-infrared (VNIR) and short-wave infrared (SWIR) ranges sensed by the multispectral imager instrument. The first satellite, S2A, is operational since 2015, followed by S2B in 2017 and recently by S2C. The presence of two satellites in the same orbit and phased at 180° gives a nominal revisit time of five days at the Equator. After atmospheric correction (L2A collection), reflectances for 12 bands (B1–9, 8A, 11, and 12) are available, with spatial resolution between 10 and 60 m, and data from S2A and S2B are considered in this work.

A key asset of MUMUCD is the inclusion of derived products based on HS data from Precursore Iperspettrale della Missione Operativa (PRISMA), launched by the Italian Space Agency and operational since 2019 [13]. Unlike Sentinel missions, its acquisition strategy is on-demand, conditioned by user priorities, weather, and geometrical constraints, and therefore, no predefined scheduling exists for a certain area. Data are collected by two HS sensors in the VNIR and SWIR ranges, with 230 nonoverlapping bands between 400 and 2500 nm, with an approximate resolution of 10 nm and a spatial resolution of 30 m. VNIR data are prioritized over those from the SWIR sensor due to their higher quality, and atmospheric-corrected (L2D) surface reflectances are processed for MUMUCD.

The dataset is further complemented with data from the Dynamic World (DW) dataset [14], providing global and near-real-time LC classes based on S2 reflectances. Available DW maps for a given location are collected around the target date, and the dominant class for each pixel is taken. Pixels not classified in the nine default DW classes (coded 0–8) are assigned to class 9 (no data). Binary change maps are then computed from bitemporal LC maps (no data pixels are conservatively set to *no change*), and further filtered over a disk with 3 pixels radius to focus on large-scale changes. Surface elevation derived from mosaicking of the digital elevation models (DEMs) from the Shuttle Radar Topography Mission (SRTM) and Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) is provided as a static field, originally available at 30-m spatial resolution.

The overall procedure followed to generate MUMUCD is illustrated in the upper portion of Fig. 1. PRISMA data are obtained from the mission data portal, while other Sentinel and ancillary datasets are available from the Google Earth Engine (GEE) platform. GEE data for Sentinel missions are gathered over a period of at least eight weeks (albeit this is typically extended for cloudy regions, such as the tropics) centered on the PRISMA target date. Least cloudiness and closeness in time to the corresponding PRISMA time of acquisition are the criteria used to select optical (average mismatch: two weeks) and SAR (average mismatch: three days) images, respectively. While the GEE platform facilitates data access and has been

TABLE I

PARAMETERS USED FOR THE GEFOLKI COREGISTRATION PROCEDURE

Modality	Radius	Level	Rank	Iteration
HS	[64,32]	3	4	4
MS	[36,18]	4	4	4
SAR	[64,32,16]	1	2	2

used in previous works for dataset generation, we note that Sentinel data over certain areas were not archived, hence corresponding pairs were discarded.

All the data are resampled from their native resolution to 10 m, the spatial resolution of S2 visible bands. Different datasets are harmonized by means of a coregistration step with the gefolki tool [15], using the S2 red band of the after image as target (from previous works [16], shifts in both S1 and S2 time series are mostly at or below pixel level). Intermediate data cubes with size 1548 × 1548 squared pixels are used in processing to reduce distortions near the edges due to coregistration. The different modalities are finally provided for the same area of interest (1536 × 1536 squared pixels), with HS data being interpolated with a nearest-neighbor approach. This preprocessing step greatly simplifies accessing and operating on the data for perspective users, allowing them to focus on their downstream applications. The coregistration procedure depends on a number of independent parameters to be determined empirically and which may be scene-dependent. Values used for MUMUCD, reported for reference in Table I, have been found by manual testing and visual inspection, with the aim of reducing coregistration artifacts (such as errors in coastal areas for optical and spurious shifts in SAR acquisitions) across different scenes, separately for optical and radar modalities. As known from previous works [17], coregistration of SAR with optical data is particularly challenging, thus a less aggressive approach was taken.

A few scenes include ephemeral changes, such as snow in the Cukotka, Eyjafjöll, Kitami, and Kirtland scenes, and these can be used to test the advantage of multiple acquisitions and multimodality. For each scene, before/after pairs for all the modalities, DEM, LC, and change maps are provided in NetCDF format. Files are provided with geographical coordinates and detailed metadata, such as information on physical units, licenses, and information on the source files (such as time of acquisition and tile). Scenes of the MUMUCD database are named based on notable geographical features (generally a town) close to or within the area of interest. A practical example of the modalities provided by MUMUCD is given in Fig. 1 (bottom) for the area of Beirut, Lebanon, showing different sensitivities of optical and radar sensing, and how bitemporal LC maps can be combined to provide an objective binary change map.

B. Related Works

With the advent of DL architectures, a number of datasets appeared to support model training. As shown in Table II, reporting a selection of datasets for CD or HS studies, the construction of these archives greatly benefited from the availability of the Sentinel data, providing near-global coverage

TABLE II

OVERVIEW OF DATASETS PROVIDING HS, MS, SAR, AND LC DATA. A † SYMBOL INDICATES MULTITEMPORAL DATASETS. THE SPATIAL RESOLUTION IS 10 M FOR ALL EXCEPT FOR HYSPECNET11K (30 M)

Name	Mode	Area (km^2)	Span
OSCD† [21]	MS	1728	4 yrs
OSCD+S1† [22]	MS/SAR	1728	4 yrs
SEN12MS [23]	MS/SAR/LC	1183986	1 yr
SEN12MS-CR-TS† [18]	MS/SAR	2544000	1 yr
BigEarthNet [24]	MS/LC	850069	1 yr
BigEarthNetMM [19]	MS/SAR/LC	850069	1 yr
HySpecNet11k [25]	HS	169323	1 wk
MUMUCD†	MS/SAR/HS/LC	33030	6 yrs

TABLE III

DESCRIPTION OF THE MAIN MODELS' SETTINGS. THE NUMBER OF CLASSES IS 2 FOR ALL THE MODELS CONSIDERED

Setting	k−means	RF	sUnet
Loss	-	Gini	NLLLoss
Estimators/weights	-	100	1.1M
N init	10	-	-
Max iterations/epochs	600	-	50

of MS and SAR acquisitions. More recently, with missions such as PRISMA and Environmental Mapping and Analysis Program (EnMAP), datasets including HS information started to appear, typically aimed at LC classification. Due to the geometry of PRISMA acquisitions (30 × 30 km^2) and the need for nearly cloud-free bitemporal acquisitions, the spatial extent of MUMUCD is smaller than datasets based on Sentinel such as SEN12MS-CR-TS [18] or BigEarthNetMM [19]. Upcoming datasets such as SpectralEarth [20] plan to further increase the scale of HS datasets, but with limited multitemporal acquisitions and without labels. In contrast, MUMUCD represents a much larger labeled dataset compared with OSCD [21], additionally with rich spectral information, thanks to the PRISMA data.

The HS datasets specifically used for CD are generally limited in terms of spatial coverage and number of acquisitions [5], [26], and the existing multimodal datasets (Table II) do not provide HS data. Compared with MS data, HS imaging provides much more detailed spectral information useful for various applications such as water and land surface characterization [3], but data are not acquired continuously, hampering their use for CD.

C. Benchmarks

We have used the MUMUCD dataset to verify the performances of widely used approaches for CD, namely, a *k*-means clustering [27] method, a random forest (RF) network [28], and a Siamese UNet (sUNet) architecture [21]. Since the clustering approach is unsupervised, we simply assign the *change* label to the minority class (based on the number of pixels), without further tuning or reliance on GT. For the sUNet case, we used nonoverlapping 32 × 32 patches, a batch size of 16, and weighted the *change* class by a factor of 5 to compensate for the strong imbalance. Selected model

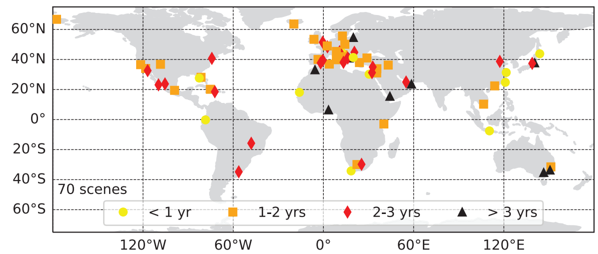

Fig. 2. Spatial distribution of the MUMUCD scenes. Markers indicate the location and the time elapsed between before/after acquisitions are color-coded.

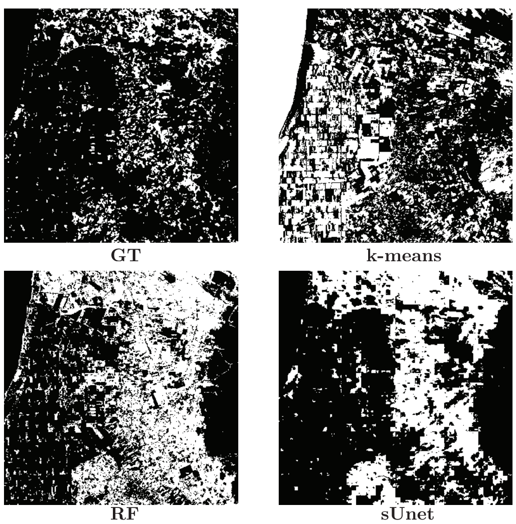

Fig. 3. CD GT (top left) and model results (with only S2 as input) for an example scene (Arborea, Italy).

hyperparameters are reported in Table III, based on sensitivity tests applied to a small MUMUCD subset. While many other models could be consider for the binary CD task, with the proposed selection we aim at covering both the ML and DL methods, with and without supervision.

To determine the advantages of multimodality, we tested the selected models under multiple input combinations: [S2] for only S2 (12 bands), [S2+ S1] (12 bands and two polarization channels), [S1+ S2+ PRS] for S1/2 and all the PRISMA bands, [S1+ (S2∩PRS)] for S1 and common bands between S2 and PRISMA, and [S1+ S2+ (PRS\S2)] for S1\2, and complementary PRISMA bands.

III. Results and Discussion

MUMUCD provides multimodal data for 70 globally distributed scenes, as illustrated in Fig. 2. The highest spatial density is found over the European region, but acquisitions in remote regions, such as the high latitudes, are included. The time between subsequent acquisitions ranges between three months and four years, with an average time span of two years.

For the results in Table IV and Fig. 3, for simplicity we split each image in 2304 patches, equally distributed among training and test splits, and trained models sequentially over all the image pairs. Other possibilities could be considered by prospective users depending on the application, for example, only considering urban scenes and assessing transfer learning capabilities by testing on unseen images only. The percentage

TABLE IV

Benchmarking of Proposed Methods for CD for Various Input Combinations. $F1$ Score for the Change Class Is Given in Percent, With Mean and Standard Deviation for the Whole Dataset

Input	k−means	RF	sUNet
S2	0.20 (0.13)	0.37 (0.15)	0.48 (0.12)
S1+S2	0.19 (0.12)	0.38 (0.15)	0.47 (0.13)
S1+S2+PRS	0.18 (0.13)	0.40 (0.15)	0.48 (0.13)
S1+(S2∩PRS)	0.18 (0.13)	0.40 (0.15)	0.48 (0.13)
S1+S2+(PRS\S2)	0.19 (0.13)	0.40 (0.15)	0.47 (0.13)

of the *change* pixels is generally 10%–15% overall, indicating how challenging the CD task can be due to the strong imbalance. The sensitivity of the different CD algorithms is shown in Table IV, with S2 being used as the main input dataset, both for comparison with previous works and since LC labels are build upon it. The results are consistent when using other metrics, such as accuracy (not shown).

As shown in Table IV, the unsupervised clustering approach gives the lower score (generally lower than 0.2), while the supervised ML method gives a notable improvement (up to 0.4). The performances of the sUNet model are superior but on the low end of those reported by [21], reaching 0.48 for multiple input combinations. This is an interesting result since, unlike the OSCD dataset, MUMUCD provides heterogeneous conditions, and therefore the model would need more generalization capabilities. The fact that relatively high scores are obtained with only S2 as input can be explained by the fact that GT is only based on DW and highlights how tailored labels should be prepared to get out the most of the multimodal collection. In practice, with the current setup, changes in the characteristics of tree-covered (e.g., stress conditions) or built (e.g., renovations) areas would be overlooked, while they would emerge with HS data [29]. Better results are to be expected with state-of-the-art networks or further model refinements. Visual inspection of the results (Fig. 3) reveals that the *k*-means method has a relatively poor match with the (unseen) GT data, independently of the provided input data. In particular, water bodies are often mistakenly assigned to the *change* class, independently of the presence of ships or other ephemeral features, as also previously noted with threshold-based methods [6]. The RF network, while trained at the pixel level, provides qualitatively better results but still tends to overestimate the *change* class fraction. We also noted that RF performances are strongly sensitive to the model hyperparameters, with a tendency of producing spurious features for shallower configurations. As anticipated from Table IV, the results for the sUNet are spatially smoother and tend to match better the GT information. The first property can be expected with convolutional networks, while the second result indicates that MUMUCD can be applied to other domains beyond urban CD (the original purpose of the sUNet).

IV. Conclusion

Motivated by the lack of global datasets leveraging the recently available HS missions, we propose the novel MUMUCD collection comprising bitemporal optical (multispectral/HS) and radar satellite acquisitions, enriched with ancillary information. This work outlines the motivation and potential uses of this ARD dataset, designed to give easy access to multimodal and multitemporal images for multiple downstream tasks. We demonstrated that despite the varied nature of MUMUCD scenes, ranging from natural to urban environments, the performances of DL models are at par with established benchmarks when using automatically defined CD GT data. While ideally annotation should be done manually to support specific applications, the inclusion of multiclass LC information would enable development of multiclass CD methods [30]. Rich metadata information and modern formats allow users to focus on the design of new architectures requiring multimodal datasets, as extensive preprocessing has been performed in advance. The dataset is available online (https://doi.org/10.5281/zenodo.10674011)

Extensions of MUMUCD could take advantage of the currently growing archives generated by HS missions beyond PRISMA, such as EnMAP by the German Space Agency [31], as well as community-contributed labels for specific applications, including innovative approaches supporting tasks such as visual queries [32].

The variety of MUMUCD makes it suitable for very different applications beyond LC CD, as prospective users may complement it with ancillary information, such as meteorological, surface geology, infrastructural. or crop type data. Owing to its large size and multiple modalities provided, this dataset can be suitable for training data-intensive foundation models [33] and serve a number of applications for societal benefit.

Acknowledgment

The information and views set out in this article are of author Le Saux and do not necessarily reflect the official opinion of the EC.

References

[1] W. A. Malila, "Change vector analysis: An approach for detecting forest changes with Landsat," in *Proc. LARS Symposia*, Jan. 1980, pp. 1–12.

[2] D. Peng, X. Liu, Y. Zhang, H. Guan, Y. Li, and L. Bruzzone, "Deep learning change detection techniques for optical remote sensing imagery: Status, perspectives and challenges," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 136, Feb. 2025, Art. no. 104282.

[3] S.-E. Qian, "Hyperspectral satellites, evolution, and development history," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7032–7056, 2021.

[4] S. Liu, D. Marinelli, L. Bruzzone, and F. Bovolo, "A review of change detection in multitemporal hyperspectral images: Current techniques, applications, and challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 140–158, Jun. 2019.

[5] S. Arjasakusuma, S. S. Kusuma, P. Melati, and A. Hafiudzan, "Change detection analysis using bitemporal PRISMA hyperspectral data: Case study of Magelang and Boyolali districts, central Java province, Indonesia," *J. Indian Soc. Remote Sens.*, vol. 50, no. 9, pp. 1803–1811, Sep. 2022.

[6] J. F. Amieva et al., "Deep-learning-based change detection with spaceborne hyperspectral PRISMA data," in *Proc. Conf. Big Data Space*, Jan. 2023, pp. 209–212. [Online]. Available: https://publications.jrc.ec.europa.eu/repository/handle/JRC135493

[7] G. Settembre, N. Taggio, N. Del Buono, F. Esposito, P. Di Lauro, and A. Aiello, "A land cover change framework analyzing wildfire-affected areas in bitemporal PRISMA hyperspectral images," *Math. Comput. Simul.*, vol. 229, pp. 855–866, Mar. 2025.

[8] S. Saha, L. Kondmann, Q. Song, and X. X. Zhu, "Change detection in hyperdimensional images using untrained models," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11029–11041, 2021.

[9] J. Nalepa et al., "Estimating soil parameters from hyperspectral images: A benchmark dataset and the outcome of the HYPERVIEW challenge," *IEEE Geosci. Remote Sens. Mag.*, vol. 12, no. 3, pp. 35–63, Sep. 2024.

[10] Z. Xiong et al., "Neural plasticity-inspired multimodal foundation model for Earth observation," 2024, *arXiv:2403.15356*.

[11] R. Torres et al., "GMES Sentinel-1 mission," *Remote Sens. Environ.*, vol. 120, pp. 9–24, May 2012.

[12] M. Drusch et al., "Sentinel-2: ESA's optical high-resolution mission for GMES operational services," *Remote Sens. Environ.*, vol. 120, pp. 25–36, May 2012.

[13] S. Cogliati et al., "The PRISMA imaging spectroscopy mission: Overview and first performance analysis," *Remote Sens. Environ.*, vol. 262, Sep. 2021, Art. no. 112499.

[14] C. F. Brown et al., "Dynamic world, near real-time global 10 m land use land cover mapping," *Sci. Data*, vol. 9, no. 1, p. 251, Jun. 2022.

[15] G. Brigot, E. Colin-Koeniguer, A. Plyer, and F. Janez, "Adaptation and evaluation of an optical flow method applied to coregistration of forest remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 7, pp. 2923–2939, Jul. 2016.

[16] A. Sebastianelli et al., "PLFM: Pixel-level merging of intermediate feature maps by disentangling and fusing spatial and temporal data for cloud removal," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5412216.

[17] L. Charrier, P. Godet, C. Rambour, F. Weissgerber, S. Erdmann, and E. C. Koeniguer, "Analysis of dense coregistration methods applied to optical and SAR time-series for ice flow estimations," in *Proc. IEEE Radar Conf. (RadarConf20)*, Sep. 2020, pp. 1–6.

[18] P. Ebel, Y. Xu, M. Schmitt, and X. X. Zhu, "SEN12MS-CR-TS: A remote-sensing data set for multimodal multitemporal cloud removal," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5222414.

[19] G. Sumbul et al., "BigEarthNet-MM: A large-scale, multimodal, multilabel benchmark archive for remote sensing image classification and retrieval [software and data sets]," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 3, pp. 174–180, Sep. 2021.

[20] N. A. A. Braham, C. M. Albrecht, J. Mairal, J. Chanussot, Y. Wang, and X. Xiang Zhu, "SpectralEarth: Training hyperspectral foundation models at scale," 2024, *arXiv:2408.08447*.

[21] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[22] S. Hafner, A. Nascetti, H. Azizpour, and Y. Ban, "Sentinel-1 and Sentinel-2 data fusion for urban change detection using a dual stream U-Net," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[23] M. Schmitt, L. H. Hughes, C. Qiu, and X. X. Zhu, "SEN12MS—A curated dataset of georeferenced multi-spectral Sentinel-1/2 imagery for deep learning and data fusion," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 4, pp. 153–160, Sep. 2019.

[24] G. Sumbul, M. Charfuelan, B. Demir, and V. Markl, "BigearthNet: A large-scale benchmark archive for remote sensing image understanding," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 5901–5904.

[25] M. H. P. Fuchs and B. Demir, "HySpecNet-11k: A large-scale hyperspectral dataset for benchmarking learning-based hyperspectral image compression methods," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2023, pp. 1779–1782.

[26] J. Lopez-Fandino, A. S. Garea, D. B. Heras, and F. Arguello, "Stacked autoencoders for multiclass change detection in hyperspectral images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 1906–1909.

[27] S. Lloyd, "Least squares quantization in PCM," *IEEE Trans. Inf. Theory*, vol. IT-28, no. 2, pp. 129–137, Mar. 1982.

[28] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.

[29] S. Vishnu, R. R. Nidamanuri, and R. Bremananth, "Spectral material mapping using hyperspectral imagery: A review of spectral matching and library search methods," *Geocarto Int.*, vol. 28, no. 2, pp. 171–190, Apr. 2013.

[30] Q. Zhu, X. Guo, Z. Li, and D. Li, "A review of multi-class change detection for satellite remote sensing imagery," *Geo-Spatial Inf. Sci.*, vol. 27, no. 1, pp. 1–15, Jan. 2024.

[31] S. Chabrillat et al., "The EnMAP spaceborne imaging spectroscopy mission: Initial scientific results two years after launch," *Remote Sens. Environ.*, vol. 315, Dec. 2024, Art. no. 114379.

[32] S. Lobry, B. Demir, and D. Tuia, "RSVQA meets bigearthnet: A new, large-scale, visual question answering dataset for remote sensing," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2021, pp. 1218–1221.

[33] C. Fibaek, L. Camilleri, A. Luyts, N. Dionelis, and B. L. Saux, "PhilEO bench: Evaluating geo-spatial foundation models," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2024, pp. 2739–2744.